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Ambunti Health Center Project
Ambunti-Drekikir District, East Sepik Province

Environmental Management Plan

1 Introduction

PDC targets the achievement of high environmental standards and strives to ensure that activities within the project area are ecologically sustainable and have minimal impact on the natural environment.

The management measures and site monitoring required to ensure that potential impacts are identified and minimized are presented in this EMP. This plan is specific to the **Construction & Completion of Ambunti Health Center at Ambunti-Drekikir District**.

The EMP is to provide:

- » PDC with a practical framework for establishing best practice environmental management standards and guidelines to mitigate potential environmental harm for each activity undertaken;
- » A mechanism to assist managers, supervisors and construction crews to comply with current legislation;
- » A means of identifying environmental issues and to provide general procedures which must be considered when undertaking construction activities;
- » A mechanism to reduce the potential impacts of the construction and operation of the facility; and
- » A basis for establishing environmental due diligence during the construction and operational phases. In essence, the EMP is to provide PDC and subcontractors with a practical guide to measure compliance by all parties with the environmental requirements. The EMP achieves this by providing a framework for comprehensive monitoring and control of construction and operation of the facility. The aim is to minimize the potential for negative environmental impact.

The EMP identifies corrective actions if monitoring indicates that the performance requirements have not been met.

The following Environmental Management Plan (EMP) details the measures to be adopted to address identified impacts during the Construction & Completion of Ambunti Health Center. The EMP details:

- » Environmental elements —the environmental aspects requiring management consideration;
- » Potential impacts —potential impacts identified in the pre-construction phase if any
- » Performance objective —the target or strategy to be achieved through management;
- » Management actions —the actions to be undertaken to achieve the performance objective, including any necessary

Approved by
Mr. Robert Akers
PROJECT MANAGER-RPHSDP

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- approvals, applications, and consultation;
- » Performance indicators —criteria against which the implementation of the actions and the level of achievement of the performance objectives will be measured;
 - » Monitoring —the intended monitoring program and the process of measuring actual performance;
 - » Responsibility —responsibility for carrying out each action is assigned to the Environment Safeguard Officer
 - » Reporting —the process and responsibility for reporting monitoring results; and

This EMP was prepared by:

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2 Management Responsibility

A number of parties have responsibilities in relation to the implementation of the EMP. All project staff have a responsibility and must adhere to the procedures outlined in the EMP at all times.

Specific management responsibilities are summarized in Table 2.

Table 2 Management responsibilities

Role	Responsibilities
Project Manager	» Implementation and monitoring of the EMP. » Provide all supervisory and management staff with an awareness and understanding of their responsibilities under this EMP. » Provide appropriate and adequate resources are allocated to allow for the effective implementation and maintenance of the EMP. » Conduct periodic reviews of environmental performance. » Report any major environmental incidents that may have a significant impact on the surrounding environment. » Provide employees and subcontractors with the relevant environmental instruction in relation to the EMP and awareness and understanding of their obligations and duties.

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Role	Responsibilities
Environment Safeguard Officer »	» Be aware of and understand the contents of and the reason for implementing the elements of the EMP and ensure all personnel including subcontractors adhere to these requirements. » Provide adequate training in the elements of the EMP to all personnel, including contractors. » Provide personnel involved in the project, including subcontractors and visitors, with the appropriate environmental training required to provide them with awareness and understanding of their responsibilities under the EMP as well as understanding of the environmental approvals that adhere to the strategies outlined in the EMP. » Carry out all work in accordance with the procedures outlined in the EMP. » Make sure that all environmental safeguards and precautions are in place and adhered to at all times at the site and activity. » Regularly inspect and monitor all activities for adherence to proper environmental safeguards. » Ensure that all equipment used is properly serviced and that all precautions are in place to prevent the likelihood of an environmental incident occurring. » Report all environmental incidents to the Superintendent as soon as practicable, but within 24 hours of them occurring.

The following Management Staff on this site are as follows:

Project Manager – Bobby Holmes

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Environmental Safeguard Officer – Anthony H. Romasanta (Full Time)

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Site Engineer (Back-up ESO) – Eduardo Delos Reyes (Full Time)

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3 Legal Requirements

During the construction phase of the project, PDC will comply with all the environmental laws, regulations and standards set by the following:

- ADB Safeguards Policy Statement
- Environment Act 2000
- Environmental Prescribed Activities Regulations
- PDC Environmental Management Plan
- Environmental Work Procedures and Guidelines

4 Special Environmental or Cultural Issues

There are no significant cultural issues for this site but there is one significant environmental risk;

- The access road is planned to enter the site on the western boundary and to proceed the full length of the boundary falling sharply in the northerly direction. This will result in the need to stabilize the road surface to prevent erosion and/or storm water runoff entering the construction areas.
- Grading and compaction of the road surface will be done prior to construction work. If needed, temporary storm water drainage system will be installed to prevent storm water runoff from entering the construction areas.

5 Scope of Works

The activities under this contract includes but not limited to the following:

- I. Mobilization / Facilities set-up.
- II. Earth works / External Works
 - Clearing & grubbing
 - Leveling/grading and trimming to subgrade
 - Excavation, backfilling & compaction to foundation level
 - Gravel road & concrete kerbs, concrete footpaths
 - Storm drainage system (spoon drains, concrete stonewall drain)
 - Perimeter security fencing & gates
 - Solar street lights, flag stand & landscaping
- III. Building Construction
 - Stage I Buildings – TB Transit House, Mother's Transit House, Outpatient & Guardian's Amenities Block, generator Shed, Cook House, Guard House, Incinerators (Electric & Rural), Maintenance Workshop, Kitchen & Laundry Block
 - Stage II Buildings – OPE and A & E Building, Family Support Center
 - Stage III Buildings – Health Center Building (Maternity and Inpatient Ward)
- IV. Rehabilitation/Demobilization
 - Dismantling of contractor facilities
 - Soil rehabilitation
 - General clean-up

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6 Plan of Works

Figure 1 – Site Plan as shown below details the boundaries, road network and layout of the buildings.

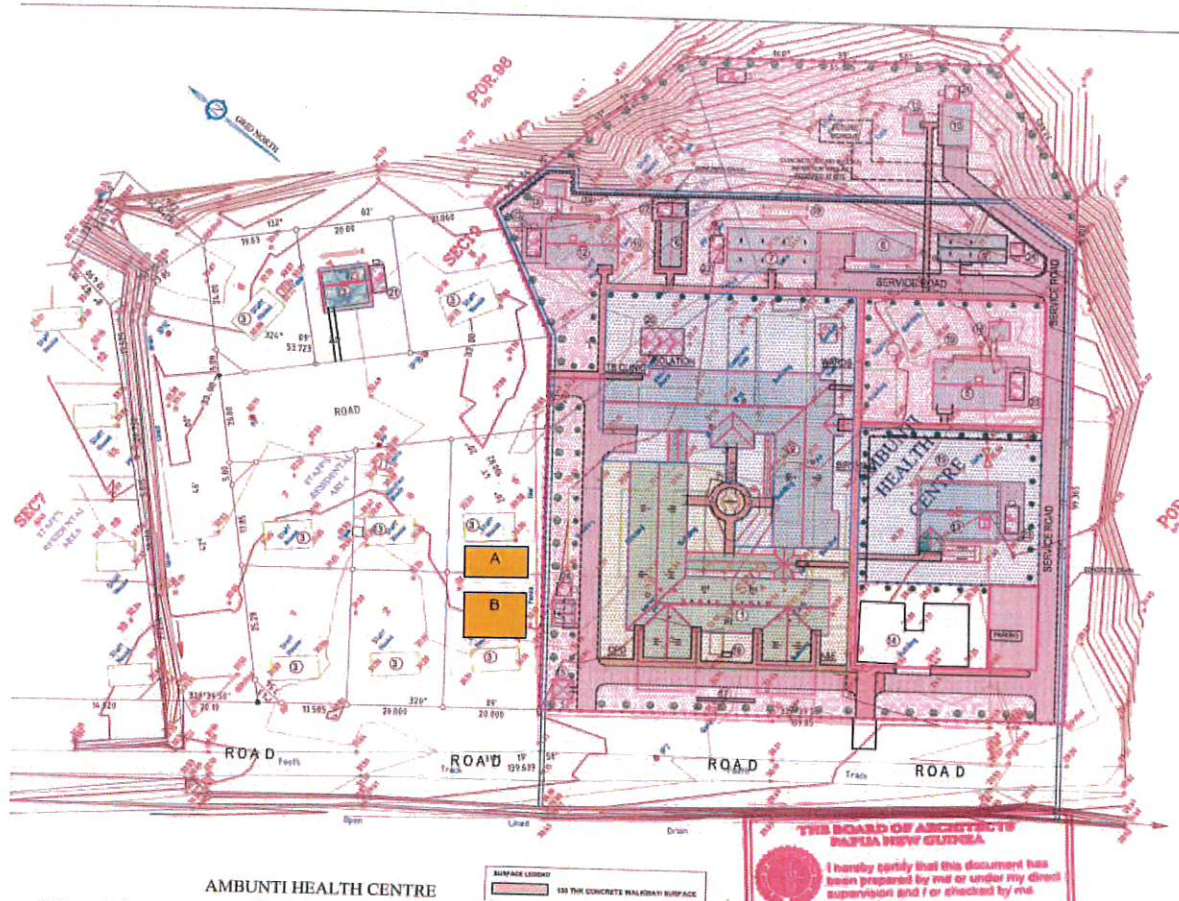
Area "A" is the proposed location of worker camp and Area "B" is the proposed location for workshops and storage facilities.

Excavation is only limited to the building footprints, road network and drainage system.

Dump site for the surplus excavated materials and waste disposal site shall be determined later on.

All aggregate materials will be shipped to the site coming from Wewak.

Figure 1 - SITE PLAN



7 Machinery and Support Equipment to be Brought to Site

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Table 1 – Machinery Table

Make and Type	Age (years)	Condition
Bulldozer	5	Good
Backhoe Loader 430E	5	Good
Power Generator (3-sets)	1	Good
12 Ton Roller	5	Good
Plate Compactors	1	Good

8 Details of Sites Used to Source Raw Materials

- All aggregates (Builder's Mix, Sand, Base Coarse) will be shipped to the site via Sepik River using tug and barge coming from the nearest source, probably Wewak.

9 Contractor's Facilities and Worker Camps

- **Contractor's Facilities (A - see Site Plan)** – A temporary workshop and storage facility will be constructed comprising of timber, corrugated roofing iron or canvass. Fuel storage area will be concrete floor and bunded. Power Generator will be the main source of electricity. It is planned to install a temporary water well as main source of water supply. However, other source for water supply will be sorted out if there is available source in the area.
- **Worker's Camp (B – see Site Plan)** – Tents will be used as worker's accommodation. Estimated number of skilled workers that will be accommodated in the camp is 30. Laborers will be hired from the local community. Gas Stoves will be used for cooking. Cooking by means of fire wood shall not be allowed.

The temporary facilities will be used in the whole duration of the project (maximum of 20 months). Upon demobilization, the temporary facilities will be dismantled accordingly and the area will be cleared and restored. All waste materials shall be disposed to the approved dump site.

Table 2 – Contractor's Facilities to be Used During Construction

	Facility	Area (m ²)	Construction			Storm water	Effluent
			Floor	Walls	Roof	Drains to...	Drains to...
1	Workers Camp	150 m ² (10mx15m)	Natural Ground	Corrugated iron	Corrugated iron	Natural drainage system	n.a.
	Kitchen & Dining	36 m ² (6mx6m)	Natural Ground	Corrugated iron (cgi)	cgi	natural drainage system	Closed septic system

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	Toilet & Shower	150 m ² (10mx15m)	concrete	Corrugate d iron (cgi)	cgi	natural drainage system	Closed septic system
	Laundry	150 m ² (10mx15m)	concrete	Corrugate d iron (cgi)	cgi	natural drainage system	Closed septic system
2	Workshop	25 m ² (5 m x 5 m)	Natural Ground	Corrugated iron (cgi)	cgi	natural drainage system	n.a.
3	machinery parking area	400 m ² (20 m x 20 m)	Natural Ground			natural drainage system	n.a.
4	Storage area – materials	400 m ² (5 m x 5 m)	Natural Ground	Corrugated iron (cgi)	cgi	natural drainage system	n.a.
5	Storage area – 200 liters' fuel drums	6.25m ² (2.5 m x 2.5 m)	Concrete bunded base			natural drainage system	n.a.

10 Environmental Management and Monitoring Plan

Table 3 – Environmental Management and Monitoring Plan

Source of Impact	Potential Impact	Mitigation and Management Plan Objective (Design Feature/Specific Measure)	Monitoring	Monitoring Frequency	Responsibility
Pre-construction Stage					
	Project will be on State owned Land with no land disputes by communities	- The existing AHC is on state owned land. The extension surveyed for the construction is also state-owned land. No landownership issues. - Land due diligence report prepared.	No Further Actions	No Further Actions	NDOH/PSU
Application for CEPA Permits	CSS and SPS not complied with	See attached approved CEPA permit.	Once – upon submission and issue	Once – upon submission and issue	PSU
Pre-construction awareness	Contractor's workers unclear of CEMP requirements	- Contractor provides induction to all workers. - Workers participate in on-site training about health and safety and general CEMP provisions.	Once before construction begins	Prior to works commencing, as new workers join.	Contractor - PDC
Construction Stage					
Vegetation clearing for construction	Planted trees, vegetation /habitat loss and fragmentation	The site is a disturbed land, with only secondary growth with planted trees. Trees and planted vegetation like, bananas, pandanus, betelnut trees, etc. are currently on state land. Clearance will be done only on surveyed and demarcated land.	Ensure that clearing is only within surveyed land.	Weekly	Contractor - PDC
	Vegetation/habitat loss and fragmentation	If practical retain large trees along the hillside slopes and base at the east and northeast side of the health centre and in front of the health facility opposite the airstrip. They provide better shade and retain landscaping within the facility.	N/A	N/A	Contractor - PDC
	Disturbance to specific ecological aspects	During site assessment, the area was identified to have only secondary regrowth and sighted no specific ecological aspects within surveyed area.	Reconfirm and Verification	Prior to disturbance	Contractor - PDC

Source of Impact	Potential Impact	Mitigation and Management Plan Objective (Design Feature/Specific Measure)	Monitoring	Monitoring Frequency	Responsibility
Topsoil removal by excavation / Destabilised landforms	Soil erosion resulting in reduced water quality during rainy seasons.	Assess and establish erosion and sediment control measures with erosion and sediment controls to be used (e.g., diversion drains, silt curtains, etc.). The controls should limit the mobilization and dispersion of sediment into nearby creeks/drainage pathways.	Assess areas of continuing erosion every week.	Ongoing	Contractor - PDC
		Where practicable, utilize land-clearing techniques that preserve the rootstock of removed vegetation in the ground.	Weekly	Ongoing	Contractor - PDC
		Where practicable, reduce disturbance of natural drainage channels during grading; avoid blocking channels with graded material.	Weekly	Ongoing	Contractor - PDC
General construction activities	Fire destroying vegetation habitats.	Leave pushed vegetation to rot at the boundary edges, rather than burning it if a fire hazard exists.	Weekly	Ongoing	Contractor - PDC
		The Sepik River is almost 400m away from the construction site, it's already disturbed by human activity, flooding carrying loads of mud/sediment and other suspended materials. However, sediment silts will still be installed to reduce sediment runoff via drains.	Weekly	Ongoing	Contractor - PDC
Runoff from cleared / disturbed areas.	Reduced water quality (particularly increased suspended solids/turbidity and in-channel sedimentation).	Site specific surface water and stormwater management procedures will be implemented.	Weekly	Ongoing	Contractor - PDC
		Stormwater runoff shall have no visible oil.	Visual Inspection	Ongoing	Contractor - PDC
		The adjoining drainage systems must be installed with sediment/silt curtains to prevent sediment into the creek.	Weekly	Ongoing	Contractor - PDC
Disposal of sewage effluent, wastewater and other liquid wastes from	Reduced water quality (particularly suspended solids/ turbidity, nutrients and	The adjoining drainage systems must be installed with sediment/silt curtains to prevent sediment into the creek.	Weekly	Ongoing	Contractor - PDC
		Design a suitable wastewater disposal strategy for sewage and grey (laundry) wastewaters such as a septic tank system.	Sampling and analysis	As agreed by Company	Contractor - PDC
		Utilize existing shower/ablation blocks at AHC to minimize sewage/laundry wastewaters.	Weekly	Ongoing	Contractor - PDC

Source of Impact	Potential Impact	Mitigation and Management Plan Objective (Design Feature/Specific Measure)	Monitoring	Monitoring Frequency	Responsibility
project-related sources	microbiological contaminants).	- Manage sewage in an appropriate manner to limit environmental contamination and protect human health. - Site specific surface water and stormwater management procedures will be implemented.	Weekly	Ongoing	Contractor - PDC
		Employ spill prevention measures by using temporary bunds, secondary containment/collection basins while refuelling, etc.	Weekly	Prior to start-up	Contractor - PDC
		Store chemicals in secure area, with concrete floor and weatherproof roof. Ensure that construction equipment and vehicles are maintained in good condition. All refuelling to be done at least 20 m away from waterways. Accidental spill action plan on site. Install sanitary toilets and washing facilities at construction site. Remove waste from site regularly for disposal to landfill. All waste disposed of as per agreed waste management plan in compliance with NDOH guidelines.	Daily	Prior to start-up	Contractor - PDC
Inappropriate waste management and disposal	Soil, groundwater, surface water and marine water contamination, ambient air quality impact	Establish appropriate processes for material, handling, storage, transportation and disposal (including recycling/reuse options), including establishment of a waste inventory and details in relation to waste minimization.	Weekly	Prior to construction and Ongoing	Contractor - PDC
		Adopt and enforce strict Duty of Care procedures, whereby a waste producer has a duty to ensure that a waste is properly managed.	Weekly	Ongoing	Contractor - PDC
		Avoid or minimize use of hazardous materials.	Weekly	Ongoing	Contractor - PDC
Hazardous materials transport, storage handling and use.	Loss of oil/fuel/other hazardous material to land, surface water, groundwater, and soil that can affect water quality and soil.	Establish site-specific hazardous material management plans to be part of the CEMP to minimize and mitigate such issues to include MSDS, training for handling, storage, use of secondary containment systems, handling procedures, etc.	Include in CEMP	Prior to Construction	Contractor - PDC
		- Use existing quarry where possible by formal agreements with resource owners in place. - Avoid extracting materials from new sites. - Have erosion/sediment control measures in place to reduce impacts on water quality. - Avoid clearing vegetation unnecessarily or too far while quarrying.	Weekly during construction	Ongoing	Contractor - PDC
Gravel and Material Extraction	Waste generation, landform deformation, erosion/sedimentation, dust nuisance, etc.				

Source of Impact	Potential Impact	Mitigation and Management Plan Objective (Design Feature/Specific Measure)	Monitoring	Monitoring Frequency	Responsibility
		• Prepare a Quarry extraction plan to include in CEMP.	Include in CEMP	Prior to Construction	Contractor - PDC
Dismantling/Re moval of Existing building structures.	Waste generation, chemical exposure from treated timber when burning, Carcinogenic from asbestos (fibro, plastic roofing, etc).	• Use the Reuse, Reduce or Recycle (3Rs) Waste Hierarchy. • The treated timber materials can be donated to community (churches, NGOs, etc) for reuse. • Materials containing asbestos must be removed with care, stored in enclosed containers for proper disposal to avoid in contact with people. • Burning of solids/debris will NOT be allowed on site.	Weekly	Ongoing during construction	Contractor - PDC
Dust from cleared / disturbed areas.	Dust emissions creating nuisance.	• Stabilize exposed areas susceptible to wind erosion using industry good practice measures such as water. • Use cleared vegetation where practicable for dust control and reinstatement.	Weekly	Ongoing	Contractor - PDC
		• Select construction equipment based on industry good practice.	Weekly	Ongoing	Contractor - PDC
Air emissions from fuel combustion and project storage.	Reduced air quality with consequent community and project nuisance	• Locate fixed and mobile equipment (e.g., generators) with consideration of local people (minimum 500 meters from residences or other sensitive receptors). • Service diesel-powered equipment regularly. • Maintain construction vehicles and equipment in order to limit emissions and remove from service any equipment from which emissions are visibly excessive.	Initial Start-up into Construction	Prior to Construction	Contractor - PDC
		• Contractor shall establish a Noise Management Plan included as part of their CEMP.	Monthly	Ongoing	Contractor - PDC
Noise emissions from construction equipment.	Intrusive and nuisance noise impacts to the local community.	• Limit construction works at night-time where practicable. • Implement noise mitigation measures (including consultation with local citizens) where practical, to reduce the noise level to acceptable levels.	Weekly	Ongoing	Contractor - PDC
			Weekly	Ongoing	Contractor - PDC

Source of Impact	Potential Impact	Mitigation and Management Plan Objective (Design Feature/Specific Measure)	Monitoring	Monitoring Frequency	Responsibility
		- Locate fixed and mobile equipment (e.g., generators) with consideration of local people (generally at a minimum distance of 500 meters from residences or other sensitive receptors). - Select construction equipment based on industry good practice. - Maintain construction vehicles and equipment in order to limit noise emissions.	Initial Start-up into Construction	Prior to Start-up	Contractor - PDC
			Include in CEMP	One-off prior to construction	Contractor - PDC
			Weekly	Ongoing	Contractor - PDC
Archaeological / Burial sites.	Fragmentation and Loss of cultural sites/values.	- Chance Finds during excavations are to be notified to CM using the Chance Find Registry Form attached in Annex 3. - Archaeological sites, burial sites discovered must be identified and demarcated to avoid destroying during construction.	Weekly	During Construction	Contractor - PDC
Public Access to Work site	Accident/Injury, exposure to hazardous materials.	- Erect barriers and warning signs around work areas. - Site accessed only by permission from Contractor. - Public awareness to people before and during construction.	Weekly	Ongoing	Contractor - PDC
Risks to worker health and safety (OHS).	Accident/Injury to Workers at Work.	- Provide personal protective equipment (PPE) to construction workers and train them in its use. - All vehicles/machinery to be properly maintained and operated in accordance with PNG road safety laws. - All loads to be secured properly.	Weekly	Ongoing during construction	Contractor - PDC
Clearance and rehabilitation of construction sites and removal of contractors' facilities		- All solid waste to be removed from sites and managed appropriately not to create additional wastes. - All contaminated soils to be removed. - All sites to be rehabilitated and restored to near-original condition. - To be included as part of final inspection before final payment is made.	Weekly	Ongoing during winding down project	Contractor - PDC

Source of Impact	Potential Impact	Mitigation and Management Plan Objective (Design Feature/Specific Measure)	Monitoring	Monitoring Frequency	Responsibility
Operational Stage					
Water Supply	Improved Water Supply from Existing Source.	As per design standard and specifications stipulated by PSU.	Monthly	During Operation	NDOH
Power Supply	Renewable and sustainable energy sources used.	As per design standard and specifications stipulated by PSU.	Monthly	During Operation	NDOH
		Environmental permits, as required, obtained.	Monthly	During Operation	NDOH
Prevention of discharge of any untreated wastewaters into the environment	Prevention of disease spread– and environmental contamination.	Sewerage systems to be built in accordance with specifications (as per the design).	Monthly	During Operation	NDOH
		All waste disposed of as per agreed waste management plan in compliance with CEPA guidelines.	Monthly	During Operation	NDOH
Correct disposal of all medical wastes	Prevention of disease spread– and environmental contamination.	Incinerators to be built in accordance with Manufacture's Specifications.	Monthly	During Operation	NDOH
		All waste disposed off as per agreed waste management plan in compliance with CEPA guidelines.	Monthly	During Operation	NDOH
Air Emissions from Incineration	Maintain ambient air quality.	Incinerator already compliant to GHG emission criteria during manufacturing and must be operated as per NDOH design Standards.	Once during installation	6 monthly if necessary	NDOH
		Checks shall be done for Ambient Air Conditions within facility boundary during operation of incinerator (if required).			

ENVIRONMENTAL WORK PROCEDURES

Issue	Performance Objective	Mitigation Measure
Access	Agreements with local land owners; Minimize vegetation clearance and erosion of exposed land surfaces	Temporary access arrangements agreed. Minimize size and duration of cleared areas. Undertake progressive re-vegetation of cleared areas.
Preparation of site (Including Contractors' facilities)	Maintain integrity of the site.	Minimize vegetative loss. Soakage areas not to discharge to surface water streams. Parking areas and workshops (if any) to have oil separators.
Septic tank installation	Minimize pollution of soil and adjacent water courses	Install as per design standard and specifications stipulated by PSU.
Gravel and material extraction	N/A. Aggregate materials will be shipped to the site coming from Wewak.	N/A. Aggregate materials will be shipped to the site coming from Kimbe.
Excavation of construction sites	Loss of topsoil	Minimize excavation area. Apply soil conservation and erosion prevention technologies. Use sediment basins. Avoid using machinery in adverse condition. Re-vegetation/protection as soon as possible.
Removal and disposal of excavated waste material (if any)	Re-use of material as much as possible	Excavated material to be stored away from site at location where it can be reused if required. Material that cannot be reused is to be landscaped so as not to cause erosion. All disposal areas to be protected to avoid erosion. All waste disposed of as per agreed waste management plan in compliance with NDOH guidelines.
Erosion and sedimentation	Minimize erosion of exposed surfaces due to high rain fall	Install sediment capture devices & Construct diversion drains to direct clean runoff away from disturbed areas. Minimize size/duration of cleared areas. Undertake progressive re-vegetation.
Storage and handling of construction materials, fuel, and lubricants	Secure storage, minimize generation of potential water pollutants, minimize accidental spills and emergency response plan in place in case accidental spills occur	Store chemicals in secure area, with concrete floor and weatherproof roof. Ensure that construction equipment and vehicles are maintained in good condition. All refueling to be done at least 20 m away from waterways. Accidental spill action plan on site. Install sanitary toilets and washing facilities at construction site. Remove waste from site regularly for disposal to landfill. All waste disposed of as per agreed waste management plan in compliance with NDOH guidelines.

Noise and vibration	Minimize nuisance to surrounding communities	Limit noisy activities to daylight hours Noise not to exceed 45 dBA at boundary of workplace.
Dust generation	Maintain air quality	If dust is carried towards residential areas or becomes problematic on site, the contractor is to apply dust control measures.
Conflict between workers and local community	Minimize friction with surrounding communities.	Any activities such as (i) use of timber/wood as fuel; (ii) hunting; (iii) clearing of areas for gardening by construction workers prohibited
Public access to site	Accident prevention	Erect barriers and warning signs around work areas Site can be accessed only by permission from Contractor
Risks to public and worker health and safety (OHS)	Minimize risk of accidents involving the public or construction workers.	Provide safety equipment to construction workers and train them in its use Secure construction site and restrict access by local community. All vehicles to be properly maintained and operated in accordance with road laws All loads to be secured properly
Use of hazardous materials	Reduction in health dangers to workers and the environment	Contractor to provide list of all hazardous chemicals/materials to be used on site. Contractor to display information sheets in work areas All such materials used and stored in compliance with NDOH guidelines
Disposal of waste materials	Prevent soil and water pollution (high water table).	All waste materials to be collected and sorted into those that can be re-used and those that need to go to an approved landfill site All waste disposed of as per agreed waste management plan in compliance with NDOH guidelines
Construction of power supply	Renewable and sustainable energy sources used	As per design standard and specifications stipulated by PSU Environmental permits, as required, obtained. No impacts on existing users (mini-hydro)
Archaeological discoveries	Prevention of the loss of cultural values	Chance discoveries are to be notified to SS
Clearance and rehabilitation of construction sites and removal of contractors' facilities	Re-established environmental amenity	All solid waste to be removed from sites and disposed of in approved landfills. All contaminated soils to be removed. All sites to be rehabilitated and restored to near-original condition. To be included as part of final inspection before final payment is made.

Table 4 – Environmental Monitoring Protocol

Receptor	Location	Monitoring Mechanism
Noise levels	Construction areas (when using machineries/equipments)	For Background Noise levels (using Noise meters if available)
Physical Infrastructure	Project Site	1. Storm Water runoff points 2. Septic Tank and Soak Pits of adequate size and presence of bunds around the pit 3. Suitable covering of soak pit to arrest fly and rodent nuisance 4. Impervious flooring in fuel storage and refuelling areas
Solid Waste Generation	Project Site	Waste segregation and disposal as per waste management plan
Fuel & Oil Wastes and Spills	Project Site	1. Storage facility, reuse and safe disposal 2. Facility or plan to control accidental oil spill 3. Availability of shovels, sands or soil 4. Presence of oil drip pans in potential oil leakages areas
Soil Erosion	Project Site	Visual observations
Socio Economics	Project Site	Local people recruited for all manual labor and other job for which local skill are available
Land	Project Site	Restoration and rehabilitation through: 1. Treatment and disposal of waste products, solids and liquid in accordance with regulatory requirements 2. Backfilling and closing of all on site disposal pits as per legal requirements

11 Staff and Worker Training

PDC will make sure that all employees and subcontractors involved with the Project receive environmental instruction in relation to the EMP.

Each person will be made aware and have an understanding of their obligations and duties detailed in this EMP.

12 Reporting

The Environment Safeguard Officer will ensure that the relevant regulatory agencies are informed as soon as possible of any incident but within 24 hours, of the incident occurring. A monthly EMP report will also be submitted during the course of construction. The monthly report will address the following:

- I. Status of work program: work completed, construction underway, and work planned
- II. Environmental unit and staff situation for the month
- III. Staff and worker awareness training carried out
- IV. Waste volumes, types, and disposal (inorganic and organic)
- V. Areas re-vegetated and rehabilitated

- VI. Dust control report
- VII. Discovery of artifacts
- VIII. Safety and monthly accident report
- IX. Status of CEMP environmental mitigation measures
- X. PSU notices issued and status of all nonconforming work
- XI. Environmental Incidents
- XII. Complaints received (as per GRM)
- XIII. Other relevant environmental issues

13 Grievance Redress Mechanism (GRM)

PDC will establish a Grievance Redress Mechanism to address project related concerns that may arise during the construction of the project. PDC will consult the local community to select a trusted literate member to be the Grievance Registrar which will hold and administer the Grievance Registration Book.

Project Name: Construction & Completion of Ambunti Health Center
Location: Ambunti-Drekikir District, East Sepik Province

The Health Services Sector Development Project welcomes complaints, suggestions, comments and queries regarding project implementation and its stakeholders. We encourage persons with grievances to provide their name and contact information to enable us to get in touch for clarification and feedback.

Should you choose to include your personal details but want that information to remain confidential, please inform us by writing/typing "(CONFIDENTIAL)" above your name.

Thank you.

Contact Information:			
Name		Gender	☐ Male
			☐ Female
Location/Address		Age	
		Phone No.	
Province		Email	
Complaint/Suggestion/Comment/Question. Please provide the details(who, what, where and how) of your grievance below: 			

How do you want us to reach you for feedback or update on your comment/grievance?

Portion to be filled by the staff:

Date received:	
Received through:	☐ in person ☐ mail ☐ email ☐ fax ☐ phone ☐ sms
Name of staff who received comment/ complaint	
Position of staff:	
Type of grievance:	
Remarks	
Signature of staff	

Update on the case:

Date	Update



**CONSERVATION AND ENVIRONMENT PROTECTION AUTHORITY
OFFICE OF THE MANAGING DIRECTOR**

7TH Floor-Dynasty Tower I
Savannah Heights, Waigani
P O Box 6601
BOROKO, NCD
Papua New Guinea

Telephone: (675) 3014500
Facsimile: (675) 3250182
E-mail: officesec@dec.gov.pg
Website: www.dec.gov.pg

The Chief Executive Officer
East Sepik Provincial Health Authority
P.O. Box 395 WEWAK,
East Sepik Province.

Date: 23rd January 2020
Our Reference: EP-L2(724)
Action Officer: AAnjo

Dear Sir/Madam,

Subject: Grant of Environment (New) Permit for Ambunti Community Health Centre Project.

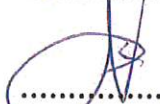
Attached please find the Environment (New) Permit **EP-L2(724)** for the development of the Ambunti Community Health Centre by the East Sepik Provincial Health Authority, approved by the Director of Environment on 23rd January 2020.

The conditions in this permit are to ensure that your activity complied with the Environment Act 2000 (amended), and its policy objectives. These conditions would be one of the reference points from which your performance will be measured against. Therefore, it is important that you should always be conscious of the Permit conditions in order to comply.

Further to the approval of the Environment Permit, your company is also hereby reminded of its obligations under this approval. Please note that we are now obligated to pay any fees owed to CEPA under this Permit. Kindly find attached the Annual Fee Notices for the outstanding annual charges owed to CEPA. You will be further informed accordingly of any other annual charges in due course.

For any clarification on the terms and conditions of the permit, please contact Mr. Anderson Anjo, Assessment Officer or the undersigned on Phone 3014500 or email: aanjo379@gmail.com.

Yours sincerely,


.....
Gunther JOKU
Managing Director



THE INDEPENDENT STATE OF PAPUA NEW GUINEA

Environment Act 2000

*Act, Sec. 65 & 66
Reg., Sec. 16(1)*

ENVIRONMENT PERMIT

An Environment Permit is hereby issued under *Section 65 of the Environment Act 2000*

TO: East Sepik Provincial Health Authority

OF: P. O. Box 395 WEWAK, East Sepik, Papua New Guinea.

- (a) to carry out upgrade and maintenance works for the Ambunti District Health Centre in Ambunti ("premises") in Ambunti Dreikikir District of East Sepik Province,
- (b) to discharge wastes into the environment from the premises while carrying out clinical works and to discharge waste whilst performing **Level 2 Activity under subcategory Level 11.4** doing Incineration, reprocessing, treatment or disposal of industrial or biomedical waste of a capacity greater than 10 tonnes per year, and
- (c) to extract water from nearby mountain sources (Yakorowai) while carrying out clinical works in the District Hospital Health Centre.

The activity shall be carried out in accordance with the terms and conditions of this permit.

Permit Number: EP-L2(724)

Date of Issue: 23rd January, 2020

Date of Commencement: 23rd January 2020

Term of Permit: Twenty-Five (25) years

Date of Expiry: 22nd January 2040



INTERPRETATIONS

Biomedical Waste: means solid and liquid waste generated during the diagnosis, testing, treatment, clinical works, research or production of biological products for humans or animals. Biomedical waste includes syringes, live vaccines, laboratory samples, body parts, bodily fluids and waste, sharp needles, cultures and lancets.

Associated Facilities: means the offices, water and sewage treatment facilities, landfill sites, chemical storage sheds, workshop, power generator shed, Incinerator, and other infrastructure described in the environment permit application which are necessary for the operations of the project and are located within the premises.

CEPA: means the Conservation and Environment Protection Authority

Extraction Point (1): refers to the location where water for hospital and domestic use is extracted from Yakorowal Stream at GPS Coordinates of 0702678 Easting and 9534376 Northing.

Discharge Point (1): refers to the location where the treated domestic wastewater from the domestic waste. The discharge point will be specified using GPS as 07022410 Easting and 9533845 Northing.

Director/Director of Environment: means the Secretary or the Head of the Conservation and Environment Protection Authority. The term is used to ensure consistency with the designation provided in s.15 of the *Environment Act* and its roles and functions under s.16 of the Act.

Domestic Wastewater: means grey water and sewage.

Environment Act/ the Act: means the *Environment Act 2000* (as amended) and any Environmental Policies and Regulations made thereunder.

Environment Permit Application: refers to the Environment Permit application for Ambunti Health Centre in East Sepik Province submitted on 10th September 2019 and received for assessment on the 18th September 2019 under the *Environment Act 2000*.

Grey Water: means waste water derived from the kitchen and laundry activities and which may contain elevated levels of BODs and detergents.

Permit: means the Environment Permit issued under s.65 of the *Environment Act 2000*. The permit authorises the carrying out of works within the premises and the discharge of wastes during the construction and operational phases of the Ambunti Health Centre.

Permit Holder: refers to East Sepik Provincial health Authority.

Premises: means the Health Center Facility at Ambunti Station at coordinates of 07022410 Easting and 9533845 Northing in Ambunti Dreikikir District of East Sepik Province.

Surface water: means watercourses, streams, creeks, rivers, and all natural bodies of surface water.

Environment Management Plan: refers to a description of environment management plan for different categories of environmental monitoring program, including the sub-plans. The plan should provide quantitative and qualitative data on all environment, health and safety aspect and impact of the project activity.

Water Quality Criteria: means the criteria for water quality as specified in *Schedule 1 of the Environment (Water Quality Criteria) Regulations 2002* and those that are prescribed as a condition of this permit.

Works: means the carrying out of activities within the premises that are concerned with the construction and operations of the health facility. These activities include land preparation, construction and operation of the health facility and other associated activities as described in the environment permit application.



TERMS AND CONDITIONS OF ENVIRONMENT PERMIT

Annual Charge

The total annual charge under this permit for a L2 (Category 11.4) activity which falls under the Fee Rank of 2.1, and will be invoiced in accordance with the Conservation and Environment Protection Authority (CEPA) Management Fees Regulation 2015 (No. 792).

Conditions

General

Permit Administration

1. The Permit Holder shall carry out works associated with the construction and operational phases of the Hospital activity in accordance with the plans and specifications in the Environment Permit Application unless specified otherwise through a condition of this permit.
2. In the event of an inconsistency between the Environment Permit Application and a condition of this Permit, the condition of this Permit will apply.

Prohibited Activities

3. The Permit Holder shall ensure **NO** work is allowed within 100 meters of identified culturally significant and sacred sites / historical sites (if any) as these sites should be reserved for cultural purposes or historical areas.
4. The Permit Holder shall ensure **NO** works are carried out within the buffer zone reserved along water courses.
5. The Permit Holder shall ensure there is no hunting, collecting and gathering of nuts, fruits, and vegetation is done by non-landowners within the premises where it is prohibited, restricted or protected as per the information provided in the Environment Permit Application, prior to clearance of vegetation.

Works

6. The Permit Holder shall observe relevant requirements and guidelines whilst carrying out construction and operation of sewage treatment plants and the incinerator.

Erosion Control

7. The Permit Holder shall design and apply appropriate measures to minimize soil erosion and limit the discharge of fugitive sediment into surface waters.

Buffer Zone

8. The Permit Holder shall maintain buffer zones of 100 meters on each side of surface water whilst carrying out works during the construction and operations of the Health Centre.

Workshop, Fuel & Waste Oil Storage Facility

9. The Permit Holder shall construct workshop and storage sheds in accordance with the minimum environmental requirements in the DEC Environmental Code of Practice for Vehicle/Machinery Workshops and Petroleum (Hydrocarbons) Storage/Resale/Usage Site, DEC, 1997.

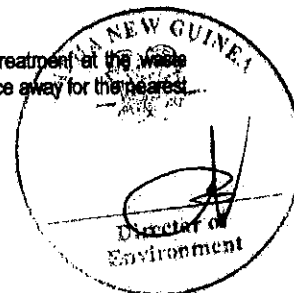
Waste Treatment Facilities

Septic Tank

10. The Permit Holder shall construct septic tanks with soak-away trench (if required) for treatment and disposal of domestic sewage and waste water from the premises.

Waste Storage

11. The Permit Holder shall identify appropriate storage locations for recyclable waste before treatment at the waste incinerator and disposal. These storage areas must be sited with a buffer zone of 100m distance away from the nearest watercourse.



Landfill

12. The Permit Holder shall construct appropriately engineered landfill site for its operations in order to securely dispose the ashes produced from the incinerator and other domestic wastes.

Waste Incinerator

13. All hazardous wastes such as biomedical wastes and oily rags are to be disposed of safely and securely off the premises. Where offsite disposal is not possible, the Permit Holder shall install an appropriate incinerator for treatment before carryout land-burial of the ashes. The incineration shall be established well away from any community dwellings and / or surface water with consideration of the prominent wind direction so as not to affect any nearby communities.
14. The Permit Holder shall ensure that the incinerator is on bunded surface which will prevent the spillage of oil, fuel, and biomedical waste that goes into the incinerator for incinerations.
15. The Permit Holder shall ensure that solidified ash from the incinerator is appropriately disposed and used according to National Health or WHO Standard and other internally accepted best practices.
16. The Permit Holder shall install air quality monitoring devices within the premises to monitor the quality of air emitted into the atmosphere.

Operations

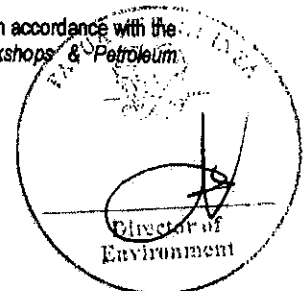
17. The Permit Holder shall observe the relevant requirements of the Water PNG Limited and any relevant guidelines during the construction and operation of sewerage treatment plant within the premises.
18. The Permit Holder shall take all reasonable measures to ensure that the waste treatment facilities are operated in accordance with the specifications and approved plans as specified in the Environment Permit Application and the approved Environment Management Plan.

Waste Treatment Facilities

19. The Permit Holder shall carry out routine inspection and maintenance of septic tank (*Condition 10*), landfill (*Condition 11*), oil-water separator (*Condition 12*), and incinerator (*Condition 13*), in order to maintain the treatment efficiency of these waste treatment facilities.
20. The Permit Holder shall ensure that leachates from the domestic wastewater effluent should not flow onto land surface from the soak-away trenches after it is discharged from the septic tanks.
21. The Permit Holder shall establish an effective awareness campaign as part of their waste management program to ensure that the downstream users and the general public are adequately informed and educated on personal health and safety aspects of waste management.
22. The Permit Holder shall ensure that the wastes generated from the hospital are securely transported for incineration or landfill site for treatment and disposal.
23. The Permit Holder shall ensure that the effluent wastewater discharged into the environment from the oil-contaminated wastewater is treated to meet the PNG Effluent Quality Standards before it is discharged into the receiving environment.
24. The Permit Holder shall ensure that the air quality emitted from this activity is clean and odourless and meets the IFC Air Quality Standards.
25. The Permit Holder shall ensure that the incinerator is environmentally friendly and the combustion process is regulated to ensure the waste materials are converted to solidified ash and flue gas.
26. The Permit Holder shall ensure that the flue gas is subjected to 850°C in the burning chamber in order to decompose the toxic gases and particulate present in it before discharging into the atmosphere.

Workshop and Fuel Storage Facilities

27. The Permit Holder shall manage the workshops and fuel storage facilities within the premises in accordance with the requirements in the DEC Environmental Code of Practice for Vehicle/Machinery Workshops & Petroleum (Hydrocarbons) Storage/Resale/Usage Site, DEC, 1997.



28. The Permit Holder shall segregate stormwater runoff from the machinery and fuel storage areas by constructing cut-off drains in order to avoid cross contamination of clean ruff with hydro-carbon wastes.

Storage of Hazardous Substances

29. The Permit Holder shall ensure that all hazardous substances (chemicals, reagents, explosives, etc. that are used within the premises are safely stored in accordance with the manufacturers Material Safety Data Sheet.

Environment Management

Environment Management Plan

30. The Permit Holder shall submit an Environment Management Plan (EMP) for the Ambunti District Hospital within three (3) months from the date of commencement of this Permit. The EMP shall incorporate all environmental, health and safety aspect and impact and incorporate the general waste management strategy of *reduce, reuse, recycle, treatment and disposal* in its treatment and disposal activities.
31. The Environment Management Plan shall include but not limited to the following sub-plans:
- I. Waste Management Plan
 - II. Landfill Management Plan
 - III. Surface and Stormwater Management plan;
 - IV. Sedimentation / Soil Erosion Control Management Plan;
 - V. Bio-medical Wastes Transportation Management Plan; and
 - VI. Incinerator Aerial Emission Management Plan.

Waste Management Plan

32. The Permit Holder shall ensure waste management sub-plan referred to in *Condition 39* contains quantitative and qualitative data on the waste streams generated within the premises and their management strategies. This plan shall incorporate the waste minimization strategy (Avoid, Reduce, Reuse, Recycle, Treatment and Disposal).

Stormwater Management Plan

33. The Permit Holder shall ensure surface and stormwater management sub-plan referred to in *Condition 39* includes a risk-based approach to monitor and manage surface water impacts to the downstream environment. This shall include:
- (i) The development of descriptive management objectives and monitoring measures to ensure that the environmental and social values of receiving surface water are maintained and protected; and
 - (ii) Maintenance of natural hydrological flows and connectivity of surface water bodies in the project area to determine the presence of any endangered species and provide information inputs to establishing site specific mitigation and management measures to protect the habitats and ecosystems processes for maintaining this critical endangered species.

Spoil/Excess Soil

34. The Permit Holder shall ensure that excess soil and unwanted rocks during construction phase are stockpiled and reused for another purpose.

Biomedical Waste

35. The Permit Holder shall ensure that all biomedical waste generated from the medical care facilities during the operation of the hospital shall be disposed correctly and safely through incineration only.

Aerial Emission

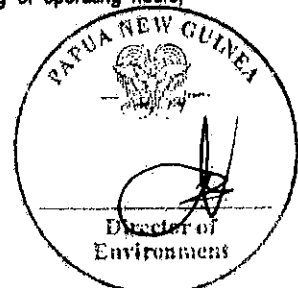
36. The Permit Holder shall ensure that the IFC General Health and Safety Guidelines and / or Australian Air Quality Standards and appropriate control measures are applied to minimize aerial emissions from the premises during the operations. Offensive odour shall not be generated from the premises.

Noise Emission

37. The Permit Holder shall ensure that noise levels within the premises are minimised through the use of appropriate noise control measures. The abatement measure may include noise barriers, scheduling of operating hours, attenuation of vehicles and equipment engines, etc.

Solid Waste

38. The Permit Holder shall dispose all domestic solid wastes appropriately at designated landfill.



39. The Permit Holder shall ensure that the heavier ash from the incinerator can be buried securely in a designated landfill.

Rainfall Runoff

40. The Permit Holder shall segregate rainfall runoff from the workshops and waste storage area by diverting clean runoff from these areas in order to avoid cross contamination with wastes.

Waste Discharge

Domestic Wastewater

41. The Discharge of Domestic wastewater at the Discharge Point (1) shall occur at the rates specified in the Table 3, below.

Table 3 - Permitted Discharge Rates for Domestic Wastewater

Description	Discharge Rate				Annual discharge volume (m ³ /yr)
	Liters per hour	Hours per day	Days per month	Days per year	
Discharge Point (1)	10	24	30	365	87.6
Operational Phase - Annual Discharge Volume					87.6 m³/year

42. The discharge of domestic waste water at the Discharge Point shall not cause the water quality within the prescribed area's water sources to exceed the water quality criteria in Table 4, below.

Table 4 - Fresh Water Quality Criteria

Parameters	Water Quality Criteria
*Faecal Coliform	≤ faecal coliform/100ml
Dissolved Oxygen	>6.0 mg/L (>80-9-% saturation)
Total Suspended Solids	< 10% change from background mean seasonal values
Oil & Grease	Non noticeable as visible film or detectable odour

Water Extraction

Water Extraction

43. The Permit Holder shall extract water at Yakorowai Stream for the domestic use and hospital use as per the rates specified in Table 1, below.

Table 1 - Permitted Water Extraction Rates for Domestic Use

Description	Extraction Rate				Annual Extraction Volume
	Liters per hour	Hours per day	Days per month	Months per year	
Extraction Point 1	12	24	30	12	103.68 m ³ /year
Total Extraction					103.68 m³/year

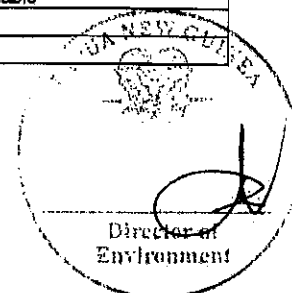
44. The Permit Holder shall ensure that the quality of the water to be used for domestic consumption complies with the *Public Health (Drinking) Water Quality Standards* in Table 2 below.

Table 2 - Water Quality Criteria for non-disinfected Water Supplies

Parameters	Highest Desirable Level	Maximum Permissible Level
Microbiological		
E. Coli (per 100ml)		None
Total Coliform (per 100ml)		<3*
Physical		
Color	5	50
Odor	Unobjectionable	Unobjectionable
Taste	Unobjectionable	Unobjectionable
Turbidity	5 units	25 units
Total Solids	500mg/L	1,500mg/L

NOTE: *Less than three Coliform organisms per 100ml if E.Coli is absent.

SOURCE: Schedule 2, *Public Health (Drinking Water) Regulations* - prepared for inclusion as at 1/1/1985.



45. The Permit Holder shall ensure that the impact on any village water supplies is considered prior to the commencement of extraction and that ongoing monitoring of supply quality, pressure or level is undertaken. Any observed changes to water quality, pressure or level shall be communicated to the Director and the local community.

Monitoring

46. The Permit Holder shall undertake monitoring of its Environmental Performance in accordance with this environment permit and approved Environment Management Plan referred to in *Condition 13*. The Monitoring shall cover all levels of impact on environment quality (i.e. water, soil, air, etc.) including noise emission and measures taken to minimize these impacts.

Reporting

47. The Permit Holder shall submit an Environmental Performance Report for the project to the Director of Environment on a biannually basis effective from the date of issuance of this permit. This report shall include but not limited to;
- (a) Raw data (analysis by certified laboratories) from the effluent and water quality monitoring program,
 - (b) Raw data from the air quality monitoring device for the operation of the Incinerator,
 - (c) Interpretation of raw data,
 - (d) Incidence of non-compliance with terms and condition of this permit and reason, and
 - (e) Status of compliance with Environmental Management Plan and other sub-Management Plans.
48. The Permit Holder shall notify the Director within 24 hours of any significant environment incidents that occur in relation to the discharge of wastes from this activity.

